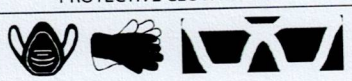
NAP+

# Material Safety Data Sheet

| NFPA Classification | DOT / TDG Pictograms | WHMIS Classification | PROTECTIVE CLOTHING |
|---|---|---|---|
| Health 1, Flammability 0, Reactivity 0, Specific Hazard |  |  |  |

## Section I. Chemical Product and Company Identification

PRODUCT NAME/ TRADE NAME: Humus Kali

SYNONYM

CHEMICAL NAME: Potassium Humate plant nutrient

CHEMICAL FAMILY: Synthetic Sequesterant.

CHEMICAL FORMULA

MATERIAL USES: Agricultural use: Fertilizer ingredient.

MSDS NUMBER: HK1

REVISION NUMBER: 01.0

MSDS prepared by the Environment, Health and Safety Department on: January 10, 2020

24 HR EMERGENCY TELEPHONE NUMBER:

Trans portation E mergency: + (662) 752-6959
Medical E mergency: + (662) 320-4100

MANUFACTURER

napnutriscience (Thailand) Company Limited
626 Bangpoo Industrial Estate, Moo 4
Praksa, Amphoe Muang, Samuthprakarn 10280,
THAILAND

DISTRIBUTOR

napnutriscience Company Limited
626 Bangpoo Industrial Estate, Moo 4
Praksa, Amphoe Muang, Samuthprakarn 10280,
THAILAND

## Section II. Hazardous Ingredients

| NAME | CAS # | Exposure Limits (ACGIH) TLV-TWA mg/m³ | TLV-TWA ppm | STEL mg/m³ | STEL ppm | CEIL mg/m³ | CEIL ppm | % by Weight |
|---|---|---|---|---|---|---|---|---|
| Potassium humate | 68514-28-3 | | | | | | | 65 Organic Matter<br>20 K2O<br>15 Humic Acid<br>13 Fulvic Acid |

TOXICOLOGICAL DATA ON INGREDIENTS

Continued on Next Page

## Section III. Hazards Identification.

| | |
|---|---|
| **POTENTIAL ACUTE HEALTH EFFECTS** | Will cause irritation of the digestive tract if swallowed. May cause severe eye irritation. May cause skin irritation. Over-exposure by inhalation may cause respiratory tract irritation. |
| **POTENTIAL CHRONIC HEALTH EFFECTS** | CARCINOGENIC EFFECTS: NONE by ACGIH, EPA, IARC, NTP, OSHA.<br>MUTAGENIC EFFECTS: NONE by ACGIH, EPA, IARC, NTP, OSHA.<br>TERATOGENIC EFFECTS: NONE by ACGIH, EPA, IARC, NTP, OSHA. |

## Section IV. First Aid Measures

| | |
|---|---|
| **EYE CONTACT** | Immediately flush eyes with water for at least 15 minutes. Use warm water if available. Get medical attention if irritation persists. |
| **MINOR SKIN CONTACT** | After skin contact wash immediately with plenty of water and non-abrasive soap. Cover irritated skin with a good quality skin cream. If irritation persists, seek medical attention. Wash contaminated clothing before reusing. |
| **EXTENSIVE SKIN CONTACT** | No additional information. |
| **MINOR INHALATION** | Allow to rest in a well ventilated area. Seek medical attention, if not feeling well. |
| **SEVERE INHALATION** | No additional information. |
| **SLIGHT INGESTION** | Irritating to the mouth, throat, stomach, digestive tract and bowels resulting in nausea, vomiting and diahrrea. Have affected person drink several glasses of water to assist in purging system. If the person is conscious, induce vomiting. Seek medical attention. |
| **EXTENSIVE INGESTION** | No additional information. |

## Section V. Fire and Explosion Data

| | |
|---|---|
| **THE PRODUCT IS** | Non-flammable. |
| **AUTO-IGNITION TEMPERATURE** | Not applicable. |
| **FLASH POINT** | Not applicable. |
| **FLAMMABILITY LIMITS** | Not applicable. |
| **PRODUCTS OF COMBUSTION** | Not applicable. |
| **FIRE HAZARD IN THE PRESENCE OF VARIOUS SUBSTANCES** | Not applicable. |
| **EXPLOSION HAZARD IN THE PRESENCE OF VARIOUS SUBSTANCES** | This substance is non-explosive. |
| **FIRE FIGHTING MEDIA AND INSTRUCTIONS** | Material will not burn. Undergoes thermal decomposition at elevated temperatures to release toxic gases. Fires involving this material may be extinguished by any means consistent with surrounding materials. As in any fire, wear MSHA/NIOSH approved pressure-demand self-contained breathing apparatus, or equivalent and full protective gear. |
| **SPECIAL REMARKS ON FIRE HAZARDS** | A self contained breathing apparatus should be used to avoid inhalation of toxic fumes. |
| **SPECIAL REMARKS ON EXPLOSION HAZARDS** | No additional information. |

***Continued on Next Page***

## Section VI. Accidental Release Measures

| | |
|---|---|
| **SPILL** | Use appropriate tools or equipment to place the spilled solid in a suitable container for reuse or disposal. Flush or wash the contaminated surface. Recover wash water and dispose of according to local and regional authority requirements. |
| | Stop leak if without risk. |

## Section VII. Handling and Storage

| | |
|---|---|
| **PRECAUTIONS** | Avoid contact with skin and eyes. DO NOT breathe dust. After handling, always wash hands thoroughly with soap and water. Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below recommended exposure limits. |
| **STORAGE** | Keep container tightly closed. Store in a dry, cool and well ventilated area. Keep away from food, drink and animal feeds. Keep out of reach of children. |

## Section VIII. Exposure Controls/Personal Protection

| | |
|---|---|
| **ENGINEERING CONTROLS** | Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below recommended exposure limits. If user operations generate dust, fumes or mist, use adequate local exhaust or general ventilation to keep exposure to airborne contaminants below the exposure limits. |
| **PERSONAL PROTECTION** | The selection of personal protective equipment varies, depending upon conditions of use. Under well controlled conditions where no direct contact with the substance exists and exposures are below the occupational exposure limit, normal work clothing may suffice. However, protective clothing and appropriate respiratory protection must be available in the area in the event of emergency. Wear appropriate respirator when ventilation is inadequate. Where skin and eye contact may occur as a result of brief periodic exposures, wear clothing with long sleeves, coverals, chemical resistant gloves, impervious chemical resistant jacket and pants, and safety glasses with side shields. A respiratory protection program that meets OSHA 29 CFR 1910.134 requirements must be followed whenever workplace conditions warrant a respirator's use. |
| **PERSONAL PROTECTION IN CASE OF LARGE RELEASE** | No further information. |
| **EXPOSURE LIMITS** | ACGIH TLV-TWA 5mg/$m^3$<br>Fed OSHA Permissible Exposure Limit: Table Z-1 8-hr Time Weighted Avg: 15 mg/$m^3$ as total dust<br>MI OSHA Permissible Exposure Limit: 10 mg/$m^3$ |

***Continued on Next Page***

Federal, State, and Provincial exposure limits may vary. Consult local officials for acceptable exposure limits in your jurisdiction.

## Section IX. Physical and Chemical Properties

| | | | |
|---|---|---|---|
| **PHYSICAL STATE AND APPEARANCE** | Powder. | | |
| **MOLECULAR WEIGHT** | Not available. | **COLOR** | Black/Dark Brown |
| **pH (1% SOLN/WATER)** | 9.5 - 10.5 | **ODOR** | Odorless |
| **BOILING POINT** | Not available. | **ODOR THRESHOLD** | Not available. |
| **MELTING POINT** | 115 C (decompose) | **TASTE** | Bitter. |
| **CRITICAL TEMPERATURE** | Not available. | **VOLATILITY** | Not available. |
| **SPECIFIC GRAVITY g/cc** | 0.80 - 0.90 | **SOLUBILITY** | Soluble in cold water, hot water, |
| **BULK DENSITY kg/m$^3$** | 800 - 900 kg/m$^3$ | **DISPERSION PROPERTIES** | Not available. |
| **VAPOR PRESSURE** | Not available. | **WATER/OIL DIST. COEFF.** | Not available. |
| **VAPOR DENSITY** | Not available. | | |

## Section X. Stability and Reactivity Data

| | |
|---|---|
| **STABILITY** | The product is stable. |
| **INSTABILITY TEMPERATURE** | Not available. |
| **CONDITIONS OF INSTABILITY** | No additional information. |
| **INCOMPATABILITY WITH VARIOUS SUBSTANCES** | Slightly reactive with moisture. |
| **CORROSIVITY** | Non corrosive |
| **SPECIAL REMARKS ON REACTIVITY** | Avoid strong oxidizing agents. |
| **SPECIAL REMARKS ON CORROSIVITY** | Contact your sales representative or metallurgical specialist to ensure compatibility with system equipment. |

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## Section XI. Toxicological Information

| | |
|---|---|
| SIGNIFICANT ROUTES OF EXPOSURE | Eye contact. Inhalation. Ingestion. |
| TOXICITY TO ANIMALS | See Section II. |
| SPECIAL REMARKS ON TOXICITY TO ANIMALS | May be harmful to livestock and wildlife if ingested. Clean up all spilled material, especially where bulk fertilizer loading of equipment occurs to prevent animal exposure. The product itself and its products of degradation are not harmful under normal conditions of use. |
| OTHER EFFECTS ON HUMANS | No additional information. |
| SPECIAL REMARKS ON CHRONIC EFFECTS ON HUMANS | No additional remark. |
| SPECIAL REMARKS ON OTHER EFFECTS ON HUMANS | No additional remark. |

## Section XII. Ecological Information

| | |
|---|---|
| ECOTOXICITY | Low toxicity for humans or animals under normal conditions of use. May be harmful to livestock and wildlife if ingested. Clean up all spilled material. |
| BOD and COD | Not available. |
| PRODUCTS OF DEGRADATION | Possibly hazardous short term degradation products are not likely. However, long term degradation products may arise. |
| TOXICITY OF THE PRODUCTS OF DEGRADATION | The products of degradation are less toxic than the product itself. |
| SPECIAL REMARKS ON THE PRODUCTS OF DEGRADATION | No additional remark.<br>Reclaiming material may not be viable. |

## Section XIII. Disposal Considerations

| | |
|---|---|
| WASTE DISPOSAL OR RECYCLING | Recover spilled material and use for intended purpose. |

## Section XIV. Transport Information

| | |
|---|---|
| DOT / TDG CLASSIFICATION | Not controlled under TDG (Canada) or D.O.T. (U.S.A.) or IATA<br>Not regulated as dangerous goods. |
| PIN and Shipping Name | Not applicable. |
| SPECIAL PROVISIONS FOR TRANSPORT | No additional remark. |
| DOT (U.S.A) (Pictograms) |  |

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## Section XV. Other Regulatory Information and Pictograms

| | | |
|---|---|---|
| **OTHER REGULATIONS** | OSHA: Hazardous by definition of Hazard Communication Standard (29 CFR 1910.1200).<br><br>TSCA (Toxic Substance Control Act): This product is listed on the TSCA Inventory. | |
| **OTHER CLASSIFICATIONS** | **HCS (U.S.A.)** | HCS: Irritating substance. May be toxic for the nervous system. |
| | **DSCL (EEC)** | R25- Toxic if swallowed.<br>R36/37/38- Irritating to eyes, respiratory system and skin. |
| **National Fire Protection Association (U.S.A.)** | Hazards presented under acute emergency conditions only: |  |
| **TDG (Pictograms - Canada)** |  | |
| **DSCL (Europe) (Pictograms)** | Not Available<br>No Disponible<br>Pas Disponible | |
| **ADR (Europe) (Pictograms)** | Not Available<br>No Disponible<br>Pas Disponible | |

## Section XVI. Other Information

**REFERENCES**

-Transportation of Dangerous Goods Act and Clear Language Regulations, current revision.
-Canada Gazette Part II, Vol. 122, No. 2 Registration SOR/88-64 31 December, 1987 Hazardous Products Act "Ingredient Disclosure List".
-Domestic Substances List, Canadian Environmental Protection Act.
-29 CFR Part 1910
-33 CFR Parts 151, 153, 154, 156
-40 CFR Parts 1-799
-46 CFR Part 153
-49 CFR Parts 1-199
-American Conference of Governmental Industrial Hygienists, Threshold Limit Values for Chemical Substances, 2003.
-NFPA 704, National Fire Codes Online, National Fire Protection Association, current edition at time of MSDS preparation.
-Corrosion Data Survey, Sixth Edition, 1985, National Association of Corrosion Engineers
-TOMES® System: Heitland G & Hurlbut KM (Eds) (electronic version): MICROMEDEX, Greenwood Village, Colorado, USA. Available at: http://csi.micromedex.com (2004). The TOMES® System includes MEDITEXT® Medical Management; HAZARDTEXT® Hazard Management; INFOTEXT® Documents; ERG2000 Emergency Response Guidebook Documents; REPROTEXT®: Heitland G & Hurlbut KM (Eds); CHRIS Hazardous Chemical Data:

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U.S. Department of Transportation, U.S. Coast Guard, Washington, D.C. (2004); HSDB: Hazardous Substances Data Bank. National Library of Medicine, Bethesda, Maryland (2004); IRIS: Integrated Risk Information System. U.S. Environmental Protection Agency, Washington, D.C. (2004); NIOSH: Pocket Guide to Chemical Hazards. National Institute for Occupational Safety and Health, Cincinnati, Ohio (2004); OHM/TADS: Oil and Hazardous Materials Technical Assistance Data System. U.S. Environmental Protection Agency, Washington, D.C. (2004); REPROTOX®: Scialli A.R. Georgetown University Medical Center and Reproductive Toxicology Center, Columbia Hospital for Women Medical Center, Washington, D.C. (2004); RTECS®: Registry of Toxic Effects of Chemical Substances. National Institute for Occupational Safety and Health, Cincinnati, Ohio (2004); and SHEPARDS: Shepard T.H.: Shepard's Catalog of Teratogenic Agents (2004).
-Michigan Office of Regulatory Reform R325.51101et seq

| | |
|---|---|
| OTHER SPECIAL CONSIDERATIONS | No additional information. |
| FOR FURTHER SAFETY, HEALTH, OR ENVIRONMENTAL INFORMATION ON THIS PRODUCT, CONTACT | napnutriscience Company Limited<br>Environment, Health and Safety Department<br>Fax (+662) 320-4100 |

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